An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

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STOC 2027 submission　(author video for the PC)

論文 v8 との差分（用語）：

- fold → flip
- notice → (newly) distinguish：同じ色の辺で真偽が食い違う。notice では「気づいてどうなるか」（そのゲートで初めて区別が生まれる）が伝わらない
- reference structure → two-witness structure $\mathcal{M}_0$ ／ read・witness edge → x-y・x-z edge
- color-determined（v8）→ color-invariant。bisimulation-invariant などと同じ「同色の点対で値が同じ」を意図

論文 v8 との差分（内容）：

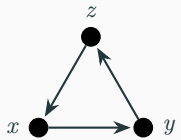
- Key Lemma「d 個を見分ける項は合成 $\geq d/2$」論文でもこの形で先に述べるとよい？
  （v8：各 fold はどこかの合成で notice される ＋ 合成 1 つで高々 2）
- distinguish を定義として明示（v8 は prediction と notice で定義）
- 省略：ループ（$(\hbar, \hbar)$ を含む）、端点 witness（p.11 で "see the paper"）。Cor 1.3（CoR 等式）、Cor 1.8・1.9（固定サイズ・乱択・ランダム入力）、項サイズ $(2 + o(1))k2^k$
- 名前のみ：Thm 1.4（FO^3 vs FO^2）、Prop 1.5（CoR vs FO^2）、Thm 1.6（ARA(3)）、Prop 1.7（RTC）は p.4 で結果の名前だけ
- 追加：導入の FO^k の例（3-cycle、推移性、closed walk）とモデル検査の表

補足：

- 投稿版の論文が確定したら定理番号を入れる（v8 の番号）：p.4 の定理 Thm 1.1、各項目 Thm 1.4・Prop 1.5・Thm 1.6・Prop 1.7、Key Lemma → Lemma 3.4・3.5、p.11 の "see the paper" → Lemma 3.5

k-variable logic FO^k

FO^k: first-order logic with k variables.
For example, in FO^3: $\varphi(x,y) := \exists z \left(E(x,y) \wedge E(y,z) \wedge E(z,x)\right)$ (triangle detection)



FO^3 also expresses, e.g.,

- transitivity: $\forall x \forall y \forall z \, (E(x,y) \wedge E(y,z) \rightarrow E(x,z))$,
- closed walk of length 4: $\exists x \exists y \left(E(x,y) \wedge \exists z \, (E(y,z) \wedge \exists y \, (E(z,y) \wedge E(y,x)))\right)$.

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

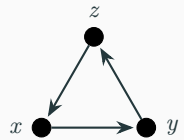
☞ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]
E.g., $O(n^3)$ for φ above: try all x, y, z

ページ概要：

- FO^k とモデル検査の問題設定
- 素直に計算すると $O(|\varphi| \cdot n^k)$

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FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under Hyperclique Hypothesis	[BS21, ICALP] [NU27, PODS]
FO^3	$2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$ under 3-OV Hypothesis	[NU27, PODS] [Wil14, CSL-LICS, for PNF] [NU27, PODS]
CoR [Tar41, JSL] Tarski's Calculus of Relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z.\ \cdots \wedge \cdots$, we use DNF translation.
$\rightsquigarrow$ $2^{O(|\varphi|)}$ compositions are used [NJ+22, JLAMP].

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- $k \geq 4$ では n^k が最適。FO^3 は特別に CoR への翻訳で n^ω まで下がる
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Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

$$\exists z.\ \big(P_1(x,z) \vee Q_1(z,y)\big) \wedge \cdots \wedge \big(P_k(x,z) \vee Q_k(z,y)\big)$$

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(This translation uses 2^k compositions, since the number of subsets $A \subseteq [k]$ is 2^k.)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])

Can we prove that there is no polynomial-size translation from FO^3 to CoR?

Partial answer [Nak22, JLAMP]: yes for **existential** CoR [Nak23, LICS] ($\equiv$ existential FO^3), via preservation under **extensions**, which fails for full CoR.

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Outline

1. Our results

2. Technical overview

3. Conclusion

Our results

The answer is **YES!**

Thm. (This paper)

There is no polynomial-size translation from FO^3 *to* CoR.
More precisely, there are FO^3 formulas $\varphi_1, \varphi_2, \ldots$ with $|\varphi_k| = O(k)$ such that every CoR term equivalent to φ_k needs $\geq 2^{k-1}$ compositions, even with shared subterms.

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$, binary (before: FO^k vs. FO^{k-1} for $k \geq 4$ [VS24, LICS])
- $\mathrm{CoR} \rightsquigarrow \mathrm{FO}^2$
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- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with RTC (reflexive-transitive closure)

Our proof technique applies to all of them!

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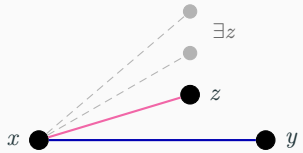
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$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z)) \right] \qquad (|G_k| = O(k))$$



$G_k(x,y)$ holds iff there is z such that
for every $i \in [k]$,
the x-y and x-z edges have labels P_i and Q_i, in either order.

CoR: 2^k **compositions** (by taking DNF, as in the example before)

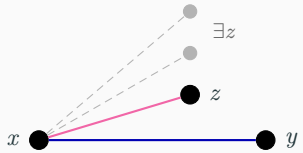
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- 下界を与える式 G_k
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Proof outline (high level)

Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

Approach We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

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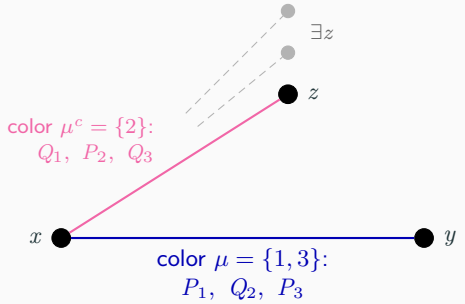
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Structures as colored complete graphs

View structures as **complete graphs**, colored with 2^k colors $\mu \subseteq [k]$: $P_i \iff i \in \mu$.

Q_i is used as the complement of P_i (i.e., $i \notin \mu$): exactly one of P_i, Q_i holds on each edge.



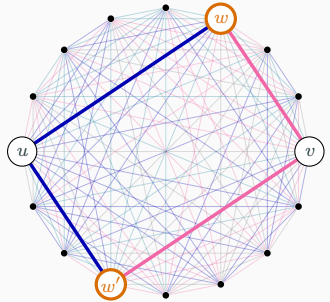
for every $i \in [3]$,
the x-y and x-z edges
have labels P_i and Q_i, in either order

☛ $G_k(x, y)$ holds iff some x-z edge has the color μ^c, where μ is the color of the x-y edge.

ページ概要：

- 構造を色付き完全グラフとして見る
- $G_k(x, y)$ は x-y の色 μ に対して、x-z の色が μ^c であるような z が存在することと同値

The two-witness structure $\mathcal{M}_0$



$\mathcal{M}_0$: $O(k\,4^k)$ vertices, constructed by, e.g., a random coloring [JMT95, Algebra Univers.]

Two-witness property For all $u \neq v$ and all colors $\boldsymbol{\mu}, \boldsymbol{\nu}$, there are at least two **witnesses** w such that the u-w and w-v edges have colors $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$, respectively.

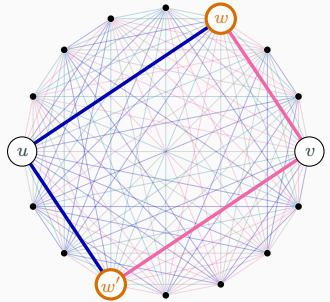
Every CoR term t is color-invariant on $\mathcal{M}_0$: $\mathcal{M}_0 \models t(u,v) \iff \mathcal{M}_0 \models t(u',v')$ whenever the u-v and u'-v' edges have the same color.

Proof by induction on t: the case $t = t_1 \,;\, t_2$ follows from the (two-)witness property.

ページ概要：

- two-witness structure $\mathcal{M}_0$（ランダム彩色で存在）
- 任意の項で同じ辺の色なら真偽が等しい（color-invariant）

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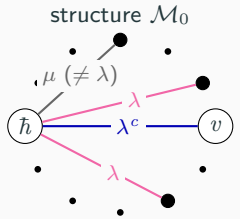
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Hub $\hbar$: a fixed arbitrary vertex of $\mathcal{M}_0$.



$\hbar$ has λ-edges

For v with the $\hbar$-v edge of color λ^c: $\mathcal{M}_0 \models G_k(\hbar, v)$ but $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$.
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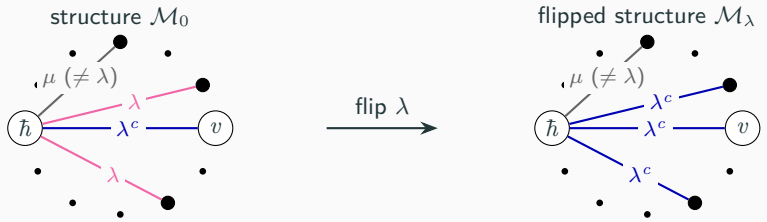
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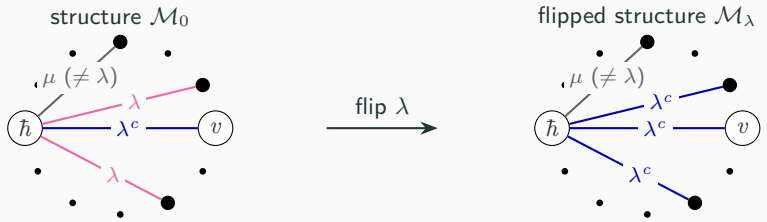
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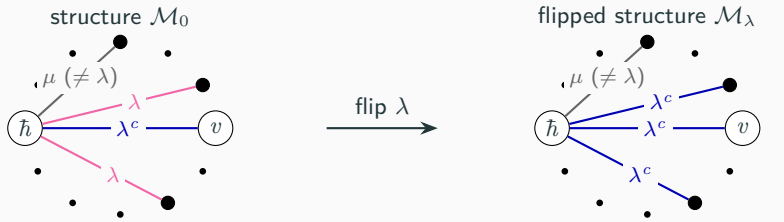
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Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (1/2)

Q. If t and s do not distinguish $\mathcal{M}_\lambda$ from $\mathcal{M}_0$, when can $t\,;s$ newly distinguish $\mathcal{M}_\lambda$?

Case $\hbar \notin \{u, v\}$: Never: $t\,;s$ cannot distinguish $\mathcal{M}_\lambda$ from $\mathcal{M}_0$ at such (u, v).

Similarly for the converse direction (from $\mathcal{M}_\lambda$ to $\mathcal{M}_0$).

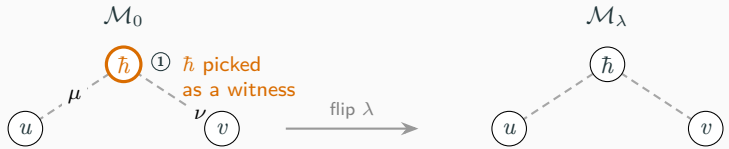
ページ概要：

- hub を含まない点対では、同じ色の組の witness を別の頂点で取れる（two-witness property）
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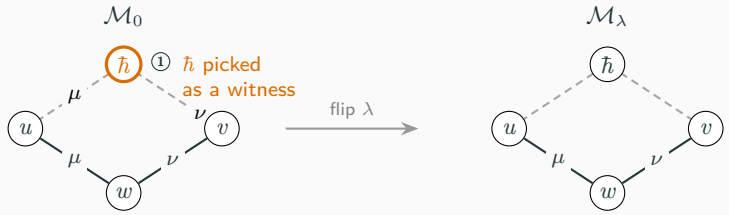
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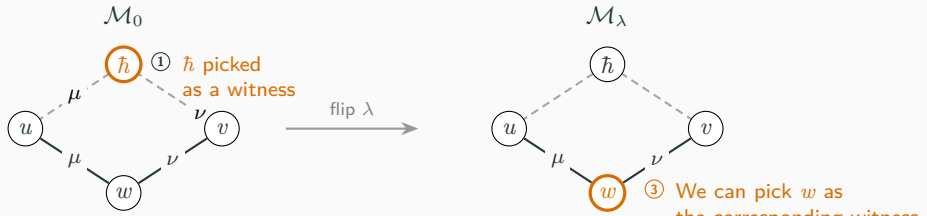
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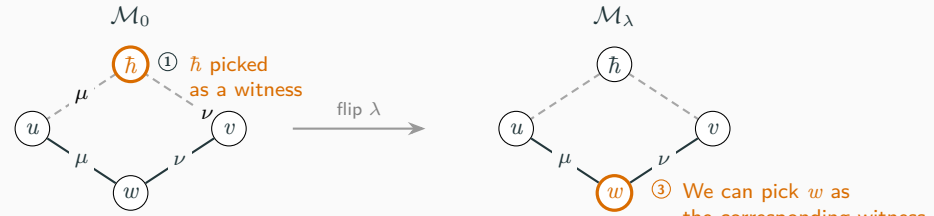
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Case $u = \hbar$ (similarly for $v = \hbar$): the $\hbar$-w edge itself may be flipped.

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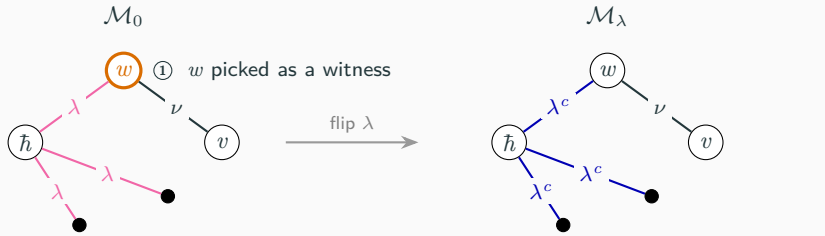
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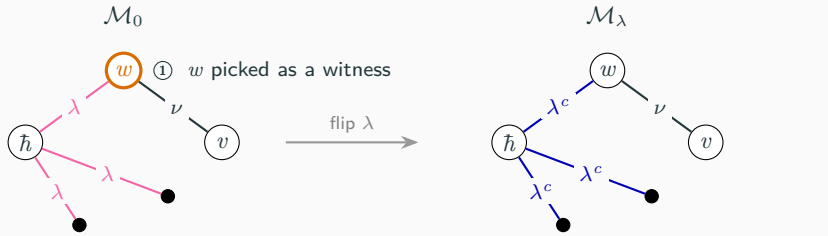
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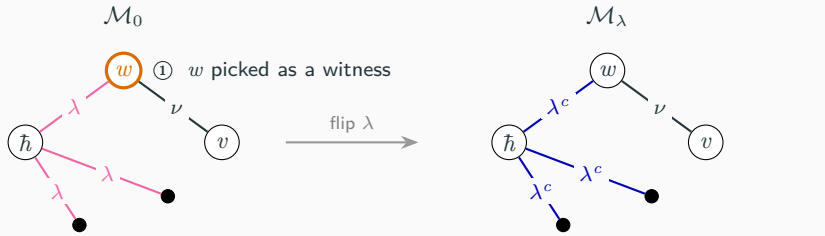
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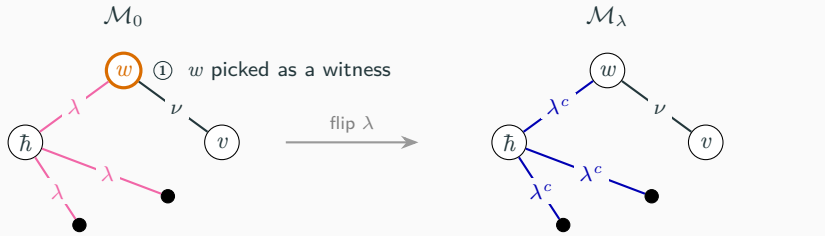
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Outline

1. Our results

2. Technical overview

3. Conclusion

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Thm.

There are FO^3 formulas $\varphi_1, \varphi_2, \ldots$ with $|\varphi_k| = O(k)$ such that every CoR term equivalent to φ_k needs $\geq 2^{k-1}$ compositions.

Our proof also applies to FO^3 vs. FO^2, matrix query languages, and CoR with RTC.

Key Idea the two-witness structure $\mathcal{M}_0$ with its 2^k flipped structures.
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Problem (Open Questions)

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Thank you for your attention!

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